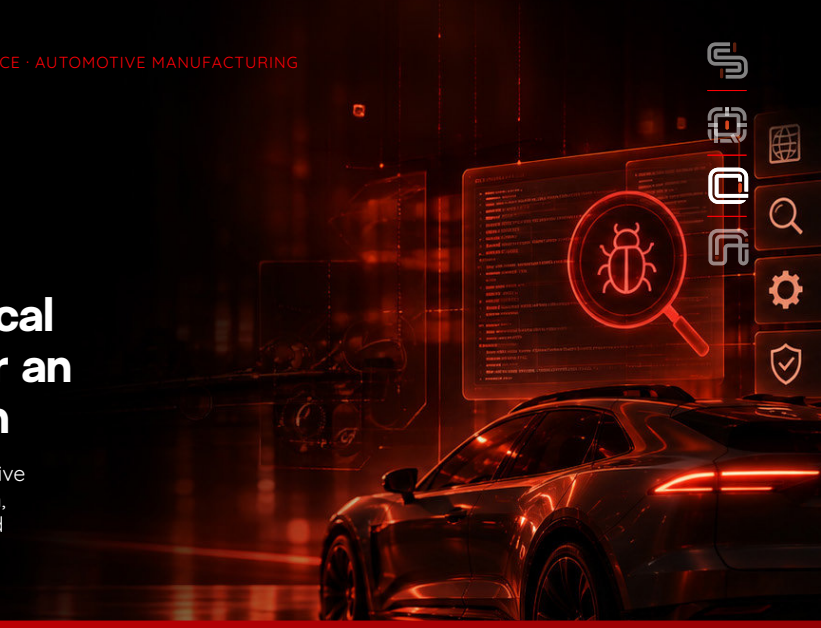
STHIROS
Strategy. Risk. Cyber. AI.

Finding and fixing 10 critical web app vulnerabilities for an automotive cloud platform

An automotive entity engaged Sthiros for comprehensive penetration testing of its cloud-hosted web application, combining black box and white box approaches to find what threat actors could exploit first.



Client	Service	Sector	Geography
An Automotive Entity	**Web Application Penetration Testing, Black Box & White Box**	**Automotive Manufacturing**	**Global**

The Challenge:

When a cloud-hosted application has never been tested from the outside in

This automotive entity needed comprehensive penetration testing of its cloud-hosted web application to identify security loopholes and vulnerabilities that could be exploited by threat actors before they were.

Authentication & Access Control

No threshold existed for failed login attempts, no captcha protected against automated attacks, and error messages disclosed sensitive information.

Session Management

Predictable session tokens exposed the application to session hijacking, with cookies missing the Secure flag and vulnerable to interception.

Security Headers & Configuration

Missing HSTS, Content-Security-Policy, anti-clickjacking, and MIME-sniffing protections left the application exposed to SSL stripping, XSS, and clickjacking attacks.

The Solution:

A five-phase testing methodology, benchmarked to industry standards

Sthiros ran a structured black box and white box penetration test, moving from reconnaissance through exploitation to a comprehensive remediation report, benchmarked against OWASP Top 10 and SANS Top 25.

Every vulnerability was verified through advanced exploitation testing before being included in the final assessment, ensuring findings reflected genuine, exploitable risk, not theoretical exposure.

The Methodology

1 Reconnaissance

Potential URLs identified for investigation, mapping the full surface area of the cloud-hosted web application.

2 Scanning

Every identified URL thoroughly scanned for vulnerabilities across authentication, session management, and configuration.

3 Analysis

Security posture determined and false positives eliminated, isolating genuine, exploitable vulnerabilities.

4 Exploitation

Advanced testing executed against every verified vulnerability to confirm real-world exploitability.

5 Reporting

A comprehensive assessment delivered with remediation recommendations mapped to OWASP Top 10 and SANS Top 25 standards.

What This Means

Assume the attacker gets in.

Headers are cheap insurance.

Fix the session, not just the login.

Outcomes

10

Vulnerabilities identified and remediated

Spanning authentication, session management, and security configuration.

2 Months

Engagement duration

From reconnaissance through to full remediation reporting.

Black & White

Testing approaches

Combined for complete internal and external vulnerability coverage.

OWASP + SANS

Standards applied

Every finding benchmarked against OWASP Top 10 and SANS Top 25.

Get in touch

Is your web application tested, or just deployed?

Speak to Sthiros about what a structured penetration test can unlock for your application security posture.

Connect with us:
sthiros.com

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